

Patalee Narasinghe

B.Sc. Engineering Honours
Department of Computer Science and Engineering
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Education

University of Moratuwa

Bachelor of Science (Hons) Computer Science and Engineering
Specialized in Data Science Engineering
Minor in Mathematics
CGPA: 3.73/4.00 (First Class) | Final Year GPA: 4.00/4.00

Aug 2022 - Present
(Expected Dec 2026)

Research Experience

Explainable Voice Analysis Framework for Speech Models

2025-Present

Final Year Research Project - University of Moratuwa

- Designing model-agnostic explainability framework for speech models (WavLM, Wav2Vec) with layer-wise and neuron-wise analysis
- Integrating probing techniques with MDL control, SHAP with Sparse Autoencoders (SAEs) to disentangle neurons for precise neuron-level causal analysis and INLP-based subspace erasure for targeted feature removal
- Developing interactive visualization module using Flask and JavaScript for model interpretability and comparison

Few-Shot Adaptation of Multimodal Foundation Models

2025

Research Project - University of Moratuwa

- Conducted comprehensive empirical study on adapting Contrastive Captioners (CoCa) for low-resource visual recognition
- Implemented and evaluated parameter-efficient fine-tuning (PEFT) methods including LoRA, hybrid prototype classification, and linear probing
- Achieved optimal results by combining Supervised Contrastive loss with cross-entropy

Teaching Assistant - Introduction to Data Science

2026

Department of Computer Science and Engineering, University of Moratuwa

- Assisted in conducting tutorial sessions for undergraduate students
- Evaluated assignments and provided feedback on data science concepts and implementations
- Mentored junior students in their research projects on machine learning applications

Publications

Causal Redundancy in Speech Representations: The Hydra Effect and Limits of Sparse Disentanglement in WavLM

2026

INTERSPEECH 2026 (Accepted)

Beyond the Black Box: A Comparative Analysis of Mechanistic and Functional Explainability in Deep Speech Models 2026

FLINS-ISKE 2026 (Accepted)

Parameter-Efficient Fine-Tuning for Vision-Language Models: The Post-Transformer Evolution 2026

ICIPRoB 2026 10.1109/ICIPRoB69625.2026.11497870

Trilingual Topic Modeling of Sri Lankan Parliamentary Debates 2026

MERCon 2026 (Pending)

A Fault of our Stars: Behavioral Drivers of Rating–Sentiment Incongruence 2026

MERCon 2026 (Pending)

Adapting Multimodal Foundation Models for Few-Shot Learning: A Comprehensive Study on Contrastive Captioners 2025

arXiv Preprint 10.48550/arXiv.2512.12824

Professional Experience

Machine Learning Engineer

Jul 2026 - Present

CML Insight

- Building agentic AI systems, causal AI models, and ETL pipelines, alongside interactive dashboards for data-driven insights
- **Technologies:** Langchain, GCP, Kubernetes, Python, Docker

Software Engineering Trainee (MLOps)

2024 - 2025

Pekoe.ai

- **ML Pipeline & Deployment:** Designed and automated end-to-end MLOps pipelines for tea quality models and image processing (12,000+ weekly images) using AWS CodePipeline, CodeBuild, Step Functions, Lambda, EventBridge, S3, ECR, and MongoDB
- **Transfer Learning:** Applied fine-tuning techniques to adapt pre-trained vision models for tea leaf feature extraction with limited labeled data
- **Infrastructure as Code:** Managed cloud resources using Terraform and configuration management with Ansible
- **Web Development:** Designed and developed frontend dashboards with react and scalable serverless backends with AWS, python, containers.
- **Technologies:** AWS (IAM , EC2, Lambda, Step Functions, S3, ECR, CodePipeline etc.), Docker, Terraform, Ansible, MongoDB, React, Node.js

Technical Skills

Research Methods	Transfer Learning, Vision & Audio, Explainable AI
Machine Learning	PyTorch, Scikit-learn, Matplotlib
MLOps & DevOps	AWS, Docker, Terraform, Ansible, CI/CD, Linux
Programming	Python, Java, C, JavaScript/TypeScript
Data Management	MongoDB, PostgreSQL, MySQL, Pandas, NumPy
Web Technologies	React, Node.js, Flask, FastAPI, HTML/CSS

Projects

Sentiment Analysis & Product Improvement Engine	<i>2024</i>
Developed full-stack NLP pipeline to extract pain points from Amazon reviews using fine-tuned BERT for classification, LSTM for emotion detection, LIME for explainability, and LLAMA 3 for actionable business recommendations.	
Technologies: Python, Hugging Face Transformers, LIME, AWS EC2 & S3, React, Flask, GitHub Actions CI/CD	
Agentic AI Report Generation Engine	<i>2024</i>
Designed serverless orchestrator using Lambda to parse JSON report schemas and dynamically dispatch content-level analysis tasks to specialized AI agents powered by AWS Bedrock, autonomously querying Knowledge Bases and SQL databases for context-aware retrieval.	
Technologies: AWS Bedrock, Lambda, AppSync, S3, Python, SQL, React	
Ecommerce platform (Web)	<i>2023</i>
Developed an Ecommerce platform for a company selling toys and electronics. This web-based application was designed to handle various functionalities such as login, registration, authorization, cart handling , sessions etc.	
Technologies: Flask, HTML, CSS, MySQL	

Certifications

Aviatrix Multi-cloud Network Associate	<i>2025</i>
Nvidia Fundamentals of Deep Learning	<i>2025</i>
AWS Data Engineering	<i>2025</i>
Kaggle Intermediate Machine Learning	<i>2024</i>

Achievements

Runners-up - IESL Techno Hackathon 2023
6th Place - SLIIT Bashway 2023
Finalist - Cypher Hackathon 2023 & 2024

References

Prof. Uthayasanker Thayasivam Supervisor - Final year Research Dept. of Computer Science and Engineering University of Moratuwa rtuthaya@cse.mrt.ac.lk	Dr. Tharindu Kaluarachchi Supervisor - Industrial Training Chief Technology Officer Pekoe (Pte) Ltd. tharindu@pekoe.ai
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